

RAKSHITH GANGADHAR MURTHY

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Education

New York University

M.Sc. in Computer Science

May 2025

New York, NY

Visveshwaraya Technological University

B.Tech. in Computer Science

Jul 2020

Bangalore, India

Experience

Smart Sentry AI

Oct 2020 – Aug 2022

Software Engineer - DevOps, Cloud & Infrastructure

Bangalore, India

- Led multicloud infrastructure across **AWS, Azure, and GCP**, managing 4 AWS accounts using WatchTower, setting up IAM roles, monitoring with CloudWatch, and private network tunneling for hybrid cloud environments
- Automated integration and deployment processes using Bitbucket CI/CD pipelines, significantly reducing manual overhead and **improving deployment speed by 40%**
- Re-architected and optimized image processing workflows by migrating from state machines to SQS queues, decreasing average image processing time from **8 seconds to 3.5 seconds**, and resulting in **\$6000/month savings**
- Deployed and managed a high-throughput image processing pipeline on AWS using Lambdas, DynamoDB, SQS, EC2 Auto Scaling, SNS, CloudWatch and API Gateway, processing over **1.5 million images per day**
- Architected a high-throughput and scalable video processing pipeline on Azure using Kubernetes, Docker, Istio, KEDA, Kafka, and MongoDB, scaling to process over **100,000 videos per day** with ELK stack observability
- Benchmarked and deployed AI/ML models, optimized GPU selection for cost-performance tradeoffs, and implemented model versioning with rollback support for production safety
- Engineered scalable **backend systems and partner integrations in Python** to support customer facing features and internal automation; mentored interns on reporting and automation tasks, enabling delivery of insights to upper management and improving reporting turnaround

NEC Technologies

Jul 2019 – Aug 2019

Full Stack Engineer Intern

Bangalore, India

- Designed and prototyped Interview Management Application to **organize and streamline interview scheduling process** using the tech stack AngularJS, NodeJS, ExpressJS, MongoDB, HTML and CSS
- Designated as the **head of the project** and tasked with daily progress reports and task assignments

Projects

MLOps Pipeline for Rodent Risk Prediction | *Python, Ansible, Terraform, OpenStack* | [GitHub](#)

- **Constructed an MLOps pipeline** to predict the likelihood of rodent infestation in top 10 restaurants in each NYC borough using GTNN and XGBoost
- Built a **Python-based ETL pipeline** to download, clean, and transform over **10GB of NYC OpenData** (311 service requests, building permits, restaurant inspections) and weather datasets as input for machine learning prediction models
- Assisted in setting up a **one-click infrastructure and services deployment** like MinIO, ArgoCD, Prometheus, Grafana, MLFlow, Ray on ChameleonCloud using OpenStack, Terraform and Ansible

Computer Vision based Rice Quality Classifier | *Python, ReactJS, NodeJS* | [GitHub](#)

- Implemented rice kernel quality analyzer using **computer vision and Mamdami Fuzzy Inference System** and generated a quality score with 93% accuracy
- Standardized the data collection process by **designing a customized photo capture setup**, ensuring high-quality acquisition of over **500 training and testing images** for model development

Self-hosted Infrastructure for Personal Cloud Services | *Intel NUC, Raspberry Pi 4, VPS*

- Configured and maintained a **self-hosted 2-node Proxmox cluster** (home server and cloud VM) running personal infrastructure including NAS, image backup, CalDAV/WebDAV, and personal websites, all reverse-proxied via **Caddy** and secured with **Authelia** for unified authentication and access control

Technical Skills

Tools: Docker, Kubernetes, Helm Charts, Nginx, Jenkins, Git, Bitbucket, Linux

Cloud: Amazon Web Services, Microsoft Azure, Google Cloud Platform

Infrastructure: DTerraform, Ansible, Apache Kafka, Redis, MongoDB, ELK Stack, Proxmox

Languages: Python, Go, Java, Bash, JavaScript